AI Interview Preparation Guide

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1. Large Language Models (LLMs)

Q1. What is a Large Language Model?

A Large Language Model (LLM) is an AI system trained on massive text datasets to understand, generate, and manipulate human language.

Q2. How are LLMs trained?

LLMs are trained in two stages:

- Pretraining on large text corpora using self-supervised learning
- Fine-tuning on specific tasks or instructions

Q3. What is prompt engineering?

Prompt engineering is the practice of designing effective inputs to get accurate and useful outputs from LLMs.

Q4. What is few-shot vs zero-shot prompting?

- **Zero-shot:** No examples provided
- **Few-shot:** A few examples included to guide the model

Q5. What is temperature in LLM generation?

Temperature controls randomness:

- Low → more deterministic
- High → more creative

Q6. What is hallucination in LLMs?

Hallucination occurs when the model generates incorrect or fabricated information.

Q7. What is instruction tuning?

Fine-tuning a model on instruction-based datasets so it can follow human commands better.

Q8. What is RLHF?

Reinforcement Learning with Human Feedback is used to align model outputs with human preferences.

Q9. What are context windows in LLMs?

The maximum amount of text (tokens) an LLM can process in a single request.

Q10. What are embeddings?

Embeddings are numerical vector representations of text used for similarity search and retrieval.

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2. Retrieval-Augmented Generation (RAG)

Q1. What is RAG and why is it used?

RAG combines retrieval of external data with LLM generation to improve accuracy and provide up-to-date information.

Q2. What problem does RAG solve?

It reduces hallucinations and allows models to access external knowledge sources.

Q3. What are embeddings in RAG pipelines?

They convert text into vectors to enable similarity-based search.

Q4. What is a vector database?

A database designed to store and efficiently search embeddings (vectors).

Q5. What are popular vector databases?

Examples include:

- Pinecone
- FAISS
- Weaviate
- Chroma

Q6. What is chunking in RAG pipelines?

Splitting large documents into smaller chunks for better retrieval accuracy.

Q7. What is semantic search?

A search technique that finds results based on meaning rather than exact keywords.

Q8. How does retrieval work in RAG?

Query → Convert to embedding → Search vector DB → Retrieve relevant documents → Pass to LLM

Q9. How do you improve RAG accuracy?

- Better chunking strategies
- Hybrid search (keyword + semantic)
- Reranking results
- High-quality data

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3. System Design & Production AI

Q1. How would you design a scalable AI system?
Use microservices architecture, load balancing, caching, and distributed computing.

Q2. How do you deploy machine learning models?
Using APIs, Docker containers, and cloud platforms like AWS, GCP, or Azure.

Q3. What is model monitoring in production?
Tracking model performance, latency, errors, and data quality.

Q4. What is model drift and data drift?

- **Data drift:** Input data changes over time
- **Model drift:** Model performance degrades due to these changes

Q5. How do you handle large-scale inference?
Through batching, caching, model optimization, and autoscaling infrastructure.

Q6. What is model versioning?
Managing different versions of models to enable tracking, rollback, and experimentation.

Q7. What are A/B tests for ML models?
Comparing two model versions in production to determine which performs better.

Q8. How do you optimize latency in AI systems?

- Use smaller or optimized models
- Cache responses
- Use efficient hardware (GPUs/TPUs)

Q9. How would you build a real-time AI service?
Using low-latency APIs, streaming pipelines, and fast inference systems.

Q10. How do you ensure reliability of AI systems in production?
Monitoring, alerting, fallback mechanisms, retries, and redundancy.

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4. ML & Deep Learning Fundamentals

Q1. What is the difference between supervised, unsupervised, and reinforcement learning?

- **Supervised:** Uses labeled data
- **Unsupervised:** Finds patterns without labels
- **Reinforcement:** Learns via rewards and penalties

Q2. What are overfitting and underfitting?

- **Overfitting:** Model memorizes training data, poor generalization
- **Underfitting:** Model fails to learn patterns

Q3. What is bias vs variance tradeoff?

- **Bias:** Error due to overly simple model
- **Variance:** Error due to model sensitivity to data

Balancing both is key for good performance.

Q4. What are training, validation, and test datasets?

- **Training:** Used to train the model
- **Validation:** Used for tuning hyperparameters
- **Test:** Used for final evaluation

Q5. What is gradient descent and its variants?
An optimization algorithm used to minimize loss by updating model parameters.
Variants include:

- SGD (Stochastic Gradient Descent)
- Mini-batch Gradient Descent
- Adam

Q6. What is backpropagation and how does it work?
An algorithm that calculates gradients and updates weights in neural networks using the chain rule.